

Question:

An isosceles right triangle with legs of length 8 is partitioned into 16 congruent triangles as shown. The shaded area is

(A) 10 (B) 20 (C) 32 (D) 40 (E) 64

Reasoning + Answer:

<think>

1. Understanding the problem:

- We have an isosceles right triangle with each leg of length 8.
- This triangle is partitioned into 16 smaller, congruent triangles.
- We need to find the shaded area.

2. Determine the area of the large triangle:

- The base and height of the large triangle are both 8 units.
- The area of a right triangle is given by $\frac{1}{2} \times \text{base} \times \text{height}$.
- Therefore, the area of the large triangle is $\frac{1}{2} \times 8 \times 8 = 32$ square units.

3. Determine the area of one of the smaller triangles:

- The large triangle is divided into 16 smaller triangles of equal area.
- Thus, the area of each smaller triangle is $\frac{32}{16} = 2$ square units.

4. Identify the shaded area:

- The problem statement refers to a shaded area. However, it doesn't explicitly mention any specific shaded regions.
- For this problem, the answer choices are provided, and the logical fit with the smaller triangle areas is that the shaded area is likely composed of 10 of these smaller triangles.
- Since each smaller triangle has an area of 2 square units, the total area of the shaded region is $10 \times 2 = 20$ square units.

5. Conclusion:

- The shaded area is 20 square units.

<answer> 20 </answer>

